



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: ENVIRONMENTAL SCIENCE [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 5 printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

Question 1

Choose the correct answers to the questions from the given options.

[16]

(Do not copy the questions, write the correct answers only)

- (i) Domestic waste water: sewage:: Industrial waste water: _____
(a) Sludge (b) Runoff (c) Effluent (d) Flow off
- (ii) A suitable strategy for controlling population growth in India is
(a) Encouraging early marriage
(b) Accelerating birth rate
(c) Implementing family planning programme
(d) Allowing people from other countries to settle in India
- (iii) A cyclone collector works on the principle of
(a) Revolution (b) Rotation (c) Gravity (d) Gravity & Rotation
- (iv) The system of waste collection used in narrow streets where a collection truck cannot enter is known as
(a) Curbside collection (b) Block collection
(c) Door to door collection (d) Community bins
- (v) Rearrange the following steps in the ascending order that depicts DT.
Step 1: Pre industrial period with high birth and death rates
Step 2: Significant drop in birth rate
Step 3: Rapid drop in death rate
Step 4: Equilibrium in birth and death rates
(a) Step 1, Step 2, Step 3, Step 4
(b) Step 2, Step 3, Step 4, Step 1
(c) Step 1, Step 3, Step 2, Step 4
(d) Step 3, Step 1, Step 2, Step 4

- (vi) If you are made the Mayor of the town, which strategy would you recommend for traffic management?
- (a) Encouraging more private vehicle
 - (b) Encouraging more signalization
 - (c) Expanding NH to connect connect different cities
 - (d) Promoting public transportation
- (vii) An alternative to wood in the paper industry is
- (a) Bagasse (b) PVS (c) PVC (d) Aluminium
- (viii) Biosphere is an example of
- (a) In situ conservation (b) Ex situ conservation
 - (c) Environmental conservation (d) Animal conservation
- (xi) Mycorrhizae is an example of
- (a) Green manure (b) Biofertiliser
 - (c) Sewage sludge (d) Humus
- (x) Study the picture given below and state its function.



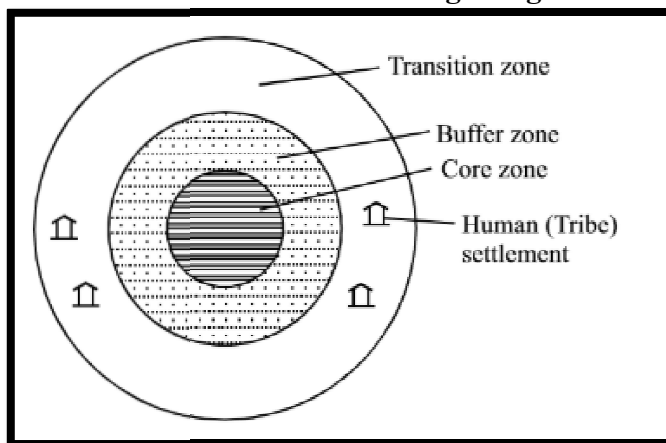
- (a) Check trade in timber
 - (b) Check trade in wildlife
 - (c) Solve trade related issues
 - (d) Works to solve disputes in MNCs
- (xi) Assertion: Crop rotation is far more effective than chemical fertilizers for enhancing soil fertility.
Reason: Crop rotation helps to put back nutrients in the soil.
- (a) A is true and R is the correct reason for A.
 - (b) A is true but R is not the correct reason for A.
 - (c) A is true but R is false.
 - (d) A is false but R is true.
- (xii) INSP promotes
- (a) Soil health (b) Human health
 - (c) Plant health (d) Animal health
- (xiii) Pick the incorrect pair from the following:
- (a) Rice husk 1. Paper making
 - (b) Kirei board 2. Make up industries

- | | |
|------------------------|-------------------|
| (c) Aluminium | 3. Wooden beams |
| (d) Polymer composites | 4. Composite wood |

(xiv) Match the following columns correctly:

<u>Column A</u>	<u>Column B</u>
(A) Pune	(i) Ralegaon Sidhhi
(B) Gravity Settling Chamber	(ii) Secondary city
(C) Anna Hazare	(iii) Hills
(D) Terrace Farming	(iv) Large particles
(a) A-i, B-iii, C-ii, D-iv	
(b) A-ii, B-iv, C-i, D-iii	
(c) A-ii, B-iii, C-iv, D-i	
(d) A-iii, B-iv, C-i, D-ii	

(xv) Identify the conservation area shown in the diagram given below.



- (a) Wildlife Sanctuary
- (b) National Park
- (c) Biosphere Reserve
- (d) Zoo

(xvi) When pollutants are emitted by checking the direction of wind to ensure that the pollutants do not move towards residential areas, the control system is known as

- (a) Zoning
- (b) Meteorological control
- (c) Tree control
- (d) Altitude control

Question 2

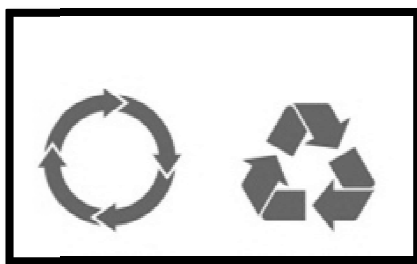
- (i) What is biodiversity hotspot? Give an example from India. [2]
- (ii) Mention any two preventive cultural practices of IPM. [2]
- (iii) What factors change the demography of a country? [2]
- (iv) Define allelopathy. How is it useful to farmers? [2]

Question 3

- (i) There are several problems associated with construction work. Mention any two environmental consequences of construction work. [2]

(ii)

[2]



What does the above picture signify? What is the difference between the two processes depicted in the picture?

(iii) What is zoning? Why is it important?

[2]

(iv) Is food aid a solution to food security? Explain your answer.

[2]

Question 4

(i) What is green manure? Why is it called so?

[2]

(ii) Mention two ways by which overgrazing can be prevented.

[2]

(iii) Write any two functions of WFP.

[2]

(iv) How is mixed farming different from mixed cropping?

[2]

SECTION B (40 MARKS)

(Attempt any four questions from this section)

Question 5

(i) What is TTZ? Why is it named so? What suggestions have been laid down by the Apex court to control environmental pollution to protect monuments?

[5]

(ii) The increasing pollution is causing imbalance between availability of resources and their utilization by the people. How can this imbalance be minimized by controlling population?

[5]

Question 6

(i) What features make Florida a sustainable city?

[5]

(ii) Plant 'A' is highly recommended for energy plantations. Give reasons for its popularity for the above mentioned purpose.

[5]

Question 7

(i)

[5]



(a) Name the type of crop shown in the above image.

(b) What are the merits and demerits of such crops?

(ii) What is conservation tillage? How does it help farmers in agriculture?

[5]

Question 8

- (i) The IUCN programme provides the framework for planning, implementing, monitoring and evaluating the conservation work undertaken by the Commissions and Secretariat with and on behalf of IUCN members. IUCN publishes the Red List. [5]
- (a) What is IUCN and Red List?
- (b) Why is Red List important?
- (ii) Mention the reasons for loss of biodiversity. [5]

Question 9

- (i) What steps can be taken to counter rural migration? [5]
- (ii) What is sewage sludge? Write its advantages and disadvantages. [5]

Question 10

- (i) What is land reform? Mention different land reform categories in India. [5]
- (ii) Identify the type of irrigation shown in the image. Why is it gaining popularity? [5]

